Projections

Orthogonal Representation

Fact. *If W is a subspace of V, then every vector $\boldsymbol{y}$ in V can be written uniquely as the sum of a vector in W and a vector in $W\perp$.*

We will call the vector in W the projection of $\boldsymbol{y}$ onto W.

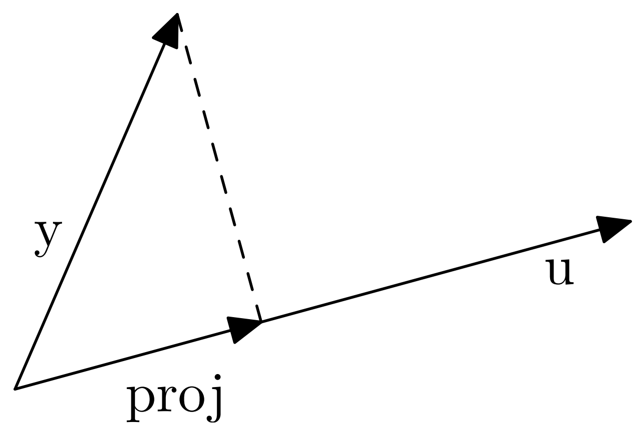
A 3-D Example

Consider in $\mathbb{R}^3$ the plane P given by $3x + 4y - z = 0$. Say we want $\boldsymbol{v} = (9, 9, 11)$ as the sum of vector in P and vector in $P^\perp$. We will see an algorithm below.

But for ad hoc approach: we know that any vector in $P^\perp$ is multiple of $\boldsymbol{w} = (3, 4, -1)$. So if we assume the requisite vector in $P^\perp$ is $a\boldsymbol{w}$, then we need $(\boldsymbol{v} - a\boldsymbol{w}) \cdot \boldsymbol{v} = 0$. This solves to $a = 2$; so $\boldsymbol{v} = 2\boldsymbol{w} + (3, 1, 13)$.

Projections

Defn. *The (orthogonal)* ***projection*** *of vector $\boldsymbol{y}$ onto vector $\boldsymbol{u}$ is its "shadow". It is denoted by* $\mathrm{proj}_{\boldsymbol{u}}(\boldsymbol{y})$.



Formula for Projection

Fact. *For vectors $\boldsymbol{y}$ and $\boldsymbol{u}$, the projection of $\boldsymbol{y}$ onto $\boldsymbol{u}$ is given by:*

$$\operatorname{proj}_{\boldsymbol{u}}(\boldsymbol{y}) = \frac{\boldsymbol{y} \cdot \boldsymbol{u}}{\boldsymbol{u} \cdot \boldsymbol{u}} \boldsymbol{u}$$

Example Projection

If $\boldsymbol{a} = (3, 4)$ **and** $\boldsymbol{b} = (-5, 2)$ **then**
$\mathrm{proj}_{\boldsymbol{b}}(\boldsymbol{a}) = (35/29, -14/29)$ **and**
$\mathrm{proj}_{\boldsymbol{a}}(\boldsymbol{b}) = (-21/25, -28/25)$**.**

Projection onto a Subspace

Defn. *The (orthogonal) projection* $\mathrm{proj}_W(\boldsymbol{y})$ *of the vector* $\boldsymbol{y}$ *onto the vector space* W *is vector in* W *such that* $\boldsymbol{y} - \mathrm{proj}_W(\boldsymbol{y})$ *in* $W^{\perp}$*.*

Fact. *If we think of* $\boldsymbol{y}$ *as a point, then the projection of it onto* W *is the closest point of* W *to it.*

Formula for Orthonormal Basis

Fact. *If W is a subspace with orthonormal basis $\{\boldsymbol{w}_i\}$, then*

$$\operatorname{proj}_W(\boldsymbol{y}) = \sum_i (\boldsymbol{y} \cdot \boldsymbol{w}_i)\, \boldsymbol{w}_i$$

Summary

If W is a subspace of V, then every vector $\boldsymbol{y}$ in V can be written uniquely as the sum of a vector in W and a vector in $W^{\perp}$. The (orthogonal) projection $\mathrm{proj}_W(\boldsymbol{y})$ of $\boldsymbol{y}$ onto W is the vector in W such that $\boldsymbol{y}-\mathrm{proj}_W(\boldsymbol{y})$ in $W^{\perp}$. Equivalently, the projection is the closest point of W to $\boldsymbol{y}$.

Summary (cont)

For vectors $\boldsymbol{y}$ and $\boldsymbol{u}$, the projection of $\boldsymbol{y}$ onto $\boldsymbol{u}$ is given by: $\mathrm{proj}_{\boldsymbol{u}}(\boldsymbol{y}) = \frac{\boldsymbol{y}\cdot\boldsymbol{u}}{\boldsymbol{u}\cdot\boldsymbol{u}}\boldsymbol{u}$. If W is a subspace with orthonormal basis $\{\boldsymbol{w}_i\}$, then $\mathrm{proj}_W(\boldsymbol{y}) = \sum_i (\boldsymbol{y}\cdot\boldsymbol{w}_i)\boldsymbol{w}_i$.